

 PREMIER PROCEDURE	TITLE: Clinical Surveillance Software Development Procedure	
	DATE CREATED: 07/06/2017	VERSION #: 4
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1. Purpose

The purpose of this Procedure is to outline the planning, processes, and templates that may be used in preparing and executing a software development program to produce commercial products developed by the Clinical Surveillance business.

2. Scope

All Clinical Surveillance Commercial Product Development programs must follow this procedure.

3. Definitions

Definitions can be found in the Clinical Surveillance Organization Acronyms and Definitions.

4. Attachments/References

Clinical Surveillance Organization Acronyms and Definitions
Clinical Surveillance Design History File Construction and Inventory
Clinical Surveillance Program Artifacts and Release Approval Matrix Procedure
Clinical Surveillance Escaped Defect Assessment and Escalation Procedure
Clinical Surveillance Software Architecture Document Procedure
Clinical Surveillance Software Design Document Procedure
Clinical Surveillance Software Verification Procedure
Clinical Surveillance Software Build/Release and Version Numbering Procedure
Clinical Surveillance Software Nonconformance and Defect Management Procedure
Clinical Surveillance Code Review and Code Inspection Procedure
Clinical Surveillance Business Features Procedure
Clinical Surveillance Usability Procedure
Clinical Surveillance Software Requirements Procedure

PROCEDURE

1. Roles and Responsibilities

Product development representatives are responsible for creating all software development plans and artifacts.

Program Team members are responsible for review and approval of the initial SDP, as well as any subsequent changes to the SDP. The Clinical Surveillance Program Artifacts and Release Approval Matrix Procedure specifies the list of approvers for the SDP.

The engineering team is responsible for the implementation of the detailed design into software code.

2. Overview

The Software Development Process for releases is based on a Phase-Gate model consisting of the following phases: Definition, Development, Qualification, and Launch.

Software Development artifacts are required for each phase and are stored in the Design History File (DHF).

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3. Process

3.1. Plan and Artifact Changes

Changes made to any plans or artifacts described in this procedure should follow the steps outlined in the Clinical Surveillance Design History File Construction and Inventory document.

3.2. Create Software Development Plan – (Definition Phase)

The completed plan dictates the details of the development program.

NOTE: No two programs are exactly alike and the plan gives the team the ability to tailor the processes and deliverables to meet the needs of the project.

Depending on program size, scope, and complexity the plans identified in section 6.2.1 below may be combined in the SDP or may be separated into individual documents and referenced in the SDP.

The SDP will be created by the Program Engineering Lead and updated for each project milestone.

The SDP shall be written in the Definition phase and stored in the DHF.

The SDP shall indicate or reference the processes, functional plans, guidelines, and checklists that will be used for each included program as well as other details relevant to the technical aspects of the program.

The SDP content may be included in the DDP document and is not required to be a stand-alone document.

3.2.1. Creating and Updating the Document

The SDP will be reviewed, updated, and routed for approval at each phase of the program as necessary.

The SDP should include pointers and/or justification for not being included, but is not limited to the following:

- Processes and methodologies used in software development
- Software configuration management plan (build/release and version numbering)
- Software nonconformance and defect management
- Software requirements plan
- Software usability plan, if applicable
- Software architecture document, if applicable
- Software design and implementation plan
- Software Tools and Resources

3.3. SDP Content

3.3.1. Create Requirements – (Definition Phase)

A Requirements plan will be specified in each program Software Development plan (SDP) in accordance to guidelines defined in the Software Requirements Procedure and the Clinical Surveillance Business Features Procedure. Examples of items to include are responsible parties creating requirements, where requirements are stored, reference to where acceptance criteria are stored, and any other information pertinent to the implementation of the requirements.

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3.3.2. Define Software Architecture - (Definition Phase)

The Software Development plan (SDP) will identify the need for architectural documentation. A summary of architectural changes will be provided in the SDP in accordance with the Clinical Surveillance Software Architecture Document Procedure. In the event that a SAD is not created, justification will be document in the SDP.

3.3.3. Usability Planning - (Definition Phase)

Planning for each program will determine the inclusion of usability considerations and will be indicated in the SDP. Usability planning will be done in accordance with the Clinical Surveillance Usability Procedure, and a reference will be included in the SDP. In the event that Usability input is not required, justification will be document in the SDP.

3.3.4. Define Software Detailed Design - (Development Phase)

The SDP for each program will outline the process for creation and reviewing software design documents which are needed to implement the software. All software design documentation will be provided as program artifacts and will conform to the Clinical Surveillance Software Design Document Procedure.

3.3.5. Tools and Resources - (Development Phase)

The SDP will capture a list of tools and resources used in the product development process.

3.3.6. Software Unit Implementation - (Development Phase)

Coding standards, design patterns, and implementation details for a specific program will be captured in the SDP.

3.3.7. Code Reviews - (Development Phase)

The code review process for each program will be detailed in the SDP. All Code reviews will be conducted in accordance with the Clinical Surveillance Code Review and Code Inspection Procedure.

3.3.8. Configuration Management - (Development Phase)

Configuration Management will be conducted in accordance with the Clinical Surveillance Software Build/Release and Version Numbering Procedure. Program details specific to the project and other specifics related to source control software will be detailed in the SDP.

3.3.9. Software Verification - (Development Phase)

Verification will be conducted in accordance with procedures defined in the Clinical Surveillance Software Verification Procedure; therefore, a verification plan is not required to be documented in the SDP.

3.3.10. Defect Tracking

The Software Defect Tracking process shall be followed in accordance with the Clinical Surveillance Software Nonconformance and Defect Management Procedure to handle defects identified during the development process. (Escaped defects identified after launch and shall be handled per the requirements of the Clinical Surveillance Escaped Defect Assessment and Escalation Procedure.)

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Description of Change

- Changed “User and Stakeholder Needs Procedure” to “Clinical Surveillance Business Features Procedure” to align with newly titled procedure.
- Updated all referenced procedure names to active hyperlinks.